Integrating Discrete-Event Simulation and Multi-Agent Reinforcement Learning for Dynamic Staff Allocation and Reduced Waiting Times in Hospitals

Dissertation submitted in part fulfillment of the requirements

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DECLARATION

I, Harishankar Somasundaram, declare that this research is my original work and that it has never been presented to any institution or university for the award of a Degree or Diploma. In addition, I have referenced correctly all literature and sources used in this work and this work is fully compliant with the Dublin Business School's academic honesty policy.

Signed: Harishankar Somasundaram
Date: January 7, 2026

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LIST OF ABBREVIATIONS

CRISP-DM	Cross Industry Standard Process for Data Mining
DES	Discrete-Event Simulation
ED	Emergency Department
ESI	Emergency Severity Index
GAE	Generalized Advantage Estimation
HCW	Healthcare Worker
ICU	Intensive Care Unit
KPI	Key Performance Indicator
LOS	Length of Stay
MADDPG	Multi-Agent Deep Deterministic Policy Gradient
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MIMIC-IV	Medical Information Mart for Intensive Care IV
RL	Reinforcement Learning
SDU	Stepdown Unit

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ABSTRACT

This study develops a novel, hybrid DES-MARL framework to dynamically allocate hospital staffs. The system was developed using a multi-agent representation of nine clinically independent departmental work groups; each represented by an autonomous agent utilizing a 17 dimensional state space that reflects staffing levels of doctors, nurses, healthcare support personnel, and administrative staff. A total of 786 patents were simulated utilizing a five stage curriculum learning strategy to progressively train two MARL algorithms (MADDPG and MAPPO) on a real-world dataset provided by MIMIC-IV. Simulations of 7 days demonstrated that MADDPG resulted in a 92.9% decrease in wait times and a 137% increase in throughput relative to baseline fixed-staffing methodologies. Additionally, the curriculum learning methodology was successful in preventing policy collapse that occurred when attempting to directly train the MARL algorithms. The proposed study represents a scalable, data-driven approach to optimize healthcare operational performance.

Keywords: Multi-Agent Reinforcement Learning, Discrete-Event Simulation, Hospital Resource Allocation, MADDPG, MAPPO, Curriculum Learning, MIMIC-IV, Healthcare Optimization

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Chapter 1

INTRODUCTION

1.1 Business Problem

Healthcare systems throughout the world are under significant pressure to provide timely and cost-effective care to a rapidly growing patient population over the last ten years. In addition to these global pressures, Irish Hospitals have been facing numerous problems with staff shortages, a steady rise in new admissions, and an aging hospital infrastructure. Reducing wait times for patients has been identified as one of the most pressing issues that will contribute to the development of high-quality and timely healthcare in Ireland.

According to Walsh and Brick (2023) there is going to be an annual need for approximately 300 more beds to support the increasing needs of Irelands aging population and its demographic growth. The number of beds needed could be up to 500 by 2040 (based on 90 percent occupancy), or as much as 1500 (based on 85 percent occupancy) based on the assumptions made in the previous statement; this highlights the urgency to implement effective strategies to optimize capacity within the current healthcare system.

Patient waiting time within a health system is impacted by a wide variety of dynamic and interrelated parameters such as staffing levels, bed usage rates, and the variable flow of patients through departments of hospitals, which Wachtel and Elalouf (2020) identified as important factors in their analysis of Emergency Department related factors. Conventional resource management techniques and linear staff scheduling approaches can handle simple deterministic and linear systems but not such complex dynamics, as proposed by El-Rifai et al. (2015) in shift scheduling using stochastic mixed-integer programming models.

This research models the behavior of hospital dynamics through Discrete-Event Simulation (DES) as reviewed by Vázquez-Serrano et al. (2021), implemented at a fine-grained temporal resolution to capture queue evolution, service completion, and patient transfers across departments. The framework enables modelling of incoming patient flow, department-level workload, staff allocation, and critical resource usage over multi-day episodes. MIMIC-IV, a publicly available database sourced from the electronic health

record of the Beth Israel Deaconess Medical Center, is used to integrate patient admissions, transfers, acuity, and care pathways into the DES (Johnson et al., 2023).

The study of these issues will be addressed through a Multi-Agent Reinforcement Learning (MARL) framework in conjunction with the DES. The proposed MARL framework was developed to meet the needs of large scale hospitals like those studied by Sivagnanam et al. (2024), and Eriksson et al. (2022) where each autonomous hospital department agent is capable of observing its local operational state as well as the overall system, and will provide continuous staffing adjustments. The proposed MARL framework will utilize two state-of-the-art MARL algorithms for coordination under partial observability, specifically MADDPG and MAPPO.

1.2 Research Problem

The research problem centres on determining optimal staff allocation strategies in hospital environments characterised by two fundamental complexities that conventional optimization approaches cannot adequately address:

1.2.1 Stochastic Complexity

Patient arrivals, treatment durations, and resource availability follow non-stationary distributions, rendering deterministic optimization ineffective (Anderson, 2014). The MIMIC-IV data used in this research demonstrated 3-4× higher complexity than synthetic simulations, with mean wait times of 1,704 minutes (28.4 hours) and mean acuity of 0.78 compared to simplified baselines.

1.2.2 Non-Linear Interdependencies

Hospital departments exhibit complex interdependencies where bottlenecks in one department cascade to affect downstream capacity. The implemented simulation models nine interconnected clinical departments where patient pathways create transfer dependencies that linear models cannot capture (Paling et al., 2020).

1.3 Aim

In order to develop, test, and validate a Discrete Event Simulation (DES) model integrated with Multi Agent Reinforcement Learning (MARL) for the purpose of optimizing hospital staff allocations in dynamic, stochastic environments with nonlinear relationships between patient demand and hospital system behavior, while minimizing patient waiting time and increasing resource utilization efficiency.

1.4 Objective

- Develop a customized DES model for modeling hospital operations in terms of stochastic patient arrivals, nonlinear service time dynamics, and patient flow between the nine clinical departments within the hospital.
- Develop a MARL model for optimizing real time hospital staff deployments, with each clinical department operating as an autonomous agent, having a 17 dimensional state space and 4 dimensional continuous action space.
- Design representations of states, actions, rewards for DES model to reflect temporal patterns, queue dynamics, and interdepartmental dependencies in order to minimize patient waiting time.
- Validate the DES/MARL model with MIMIC IV clinical data (786 processed admissions) to reflect the realities of patient demographics, acuities, and care paths.
- Conduct a comparative evaluation of MADDPG and MAPPO algorithms through a five stage curriculum learning protocol over 100 training episodes of 7 day simulations.

1.5 Hypothesis

The following hypothesis will be tested empirically:

Hypothesis 1: MARL models will achieve significantly lower patient waiting times than static staffing baselines.

Hypothesis 2: Centralized-critic architectures (MADDPG) will perform better than independent actor-critic methods (MAPPO) in environments with high levels of interdepartmental dependency because they can represent joint action-value functions.

Hypothesis 3: Curriculum learning is necessary for maintaining training stability in complex hospital systems, because direct training on difficult scenarios causes policy collapse.

These hypotheses are tested through systematic experimentation with the implemented DES-MARL framework.

1.6 Scope

The scope of this study includes the development and testing of an intelligent decision support system for dynamic hospital staffing. The following criteria will define the boundaries of this study

- **Clinical Environment:** Nine clinical departments are modeled: Emergency Department, ED Observation, Medicine, Medical/Surgical, Medicine/Cardiology, Neurology, ICU, Stepdown Unit, and Discharge Lounge.
- **Staff Categories:** Four types of personnel are evaluated: Doctors, Nurses, Healthcare Workers, and Administrative Staff with role specific capacities and cost coefficients.
- **Data Source:** The MIMIC IV clinical data set is utilized for simulating patient demographic characteristics, acuities, and care paths (786 processed admissions).
- **Algorithms:** A comparative analysis is performed on MADDPG (centralized critic, off-policy) and MAPPO (shared value function, on-policy) which represent differing paradigms of MARL.
- **Time Horizon:** Simulations span 7 days (2,016 timesteps @ 5 minute intervals) allowing for the analysis of medium term staffing strategies.
- **Training Protocol:** Curriculum learning of five stages, across 100 episodes of progressively increased difficulty.

1.7 Limitations

While the proposed framework represents advancements in the area of hospital optimization, there are several limitations that should be acknowledged:

- **Simulation Abstraction:** The DES environment abstracts from certain operational complexities of the hospital such as the dynamics of staff shift changes, skill heterogeneity of staff members, and physical layout of the hospital.
- **Single Institution Data:** Models developed utilizing MIMIC IV (Beth Israel Deaconess Medical Center) may not easily generalize to other hospitals with different patient populations or operational procedures without calibration.
- **Action Space Simplification:** The continuous action space for adjusting staffing represents the abstraction of discrete hiring/scheduling decisions and the constraints of deploying staff in practice.
- **External Factors:** The simulation does not consider external factors influencing resource availability such as seasonal disease patterns, local events, and supply chain disruptions.
- **Validation Scope:** Results are validated through simulation alone. Real world implementation will require additional clinical validation and regulatory approval.

Chapter 2

LITERATURE REVIEW

A review of the literature on which this dissertation is based is presented below. The review is thematic in nature and includes an overview of the operational challenges of hospitals; Discrete Event Simulation (DES); Reinforcement Learning (RL) in DES; Multi-Agent Systems (MAS); and the specific algorithms that were used in the research.

2.1 Healthcare System Complexity and Resource Allocation

Hospital systems have inherent complexities in terms of stochastic arrival rates of patients, varying service times, and interconnected departmental operations. Extensive documentation of the challenges associated with the optimal allocation of resources in these complex systems exists within the Operations Research literature.

Stochastic arrival rates and stochastic service times create significant complexity when attempting to optimize the efficiency of healthcare facilities by Liu et al., (2024). Emergency Departments also face unique challenges related to variability in patient arrivals and service times, which lead to inefficient resource allocation and crowding by De Santis et al., (2022). The operational challenges faced by hospitals in Ireland are well-documented by Ordu et al., (2021), and illustrate the impact of budgetary constraints on resource scarcity and the need for sophisticated optimization techniques.

The capacity crisis facing hospitals in Ireland has been quantitatively analyzed by Walsh and Brick (2023) who estimate that there will be a requirement for 300-1500 additional hospital beds by 2040. Their analysis demonstrates the need for capacity optimization techniques that allow for maximum throughput using current infrastructure.

Emergency Department waiting times are influenced primarily by staff availability and patient flow by Wachtel and Elalouf (2020) Collectively, these studies demonstrate the need for dynamic, data-driven hospital resource allocation methodologies that can dynamically adjust to changing demand patterns while maintaining consideration of operational constraints.

2.2 Discrete-Event Simulation in Healthcare

Discrete Event Simulation (DES) is a modeling technique that allows for the modeling of the dynamic interactions of patients and staff within hospitals, providing the capability to model patient flow, queueing behaviors, and resource utilization at a very granular level of detail.

Vázquez-Serrano et al. (2021) conducted a comprehensive review of DES applications in healthcare and identify common modeling techniques for patient pathways, service processes, and resource constraints. Their review establishes DES as the most commonly utilized modeling technique in hospital simulation settings, due to its ability to capture stochastic events and transition of system states.

Gunal (2012) provides practical advice regarding the construction of hospital simulation models utilizing DES and compares DES to System Dynamics and Agent-Based Simulation techniques. The trade-off between model complexity and generalizability influences the design decisions in this research, where DES was chosen for its balance of fidelity and computational tractability.

Eriksson et al. (2022) demonstrate the feasibility of integrating DES with deep reinforcement learning for scheduling problems and provide a conceptual framework that directly informs this research. Their work demonstrates the feasibility of using DES environments as training platforms for learning-based optimization.

2.3 Reinforcement Learning for Healthcare Optimization

Reinforcement learning is a mathematical framework for sequential decision-making under uncertainty, and is being increasingly applied to healthcare resource management.

Lee and Lee (2020) applied Deep Q-Networks to ED patient scheduling and demonstrated improved access for high-acuity patients although they noted limitations under peak demand conditions. Although their work demonstrates both the potential and the challenges of applying single-agent RL to hospital operations, it is illustrative of the growing interest in RL as a solution to various healthcare optimization challenges.

Lazebnik (2023) applied actor-based reinforcement learning to hospital staff and resource allocation, and demonstrated measurable improvements in treatment outcomes. However, their work notes computational complexity as a major barrier to the practical deployment of RL-based methodologies, which motivated the adoption of a curriculum learning approach in this research.

Wu et al. (2025) provide a methodological review of RL in healthcare operations and highlight four key challenges to the successful application of RL methodologies in these

settings: non-stationarity; partial observability; and credit assignment. The framework provided in Wu et al. (2025) for categorizing RL applications in healthcare settings informed the algorithmic choices in this research.

Ali (2022) provides a survey of RL applications across healthcare domains and notes that most applications rely on overly simplified assumptions (e.g., homogeneous states, stationary environments) that severely limit the applicability of these methodologies to real-world settings. This limitation to realism motivated the use of realistic MIMIC-IV data in this research.

2.4 Multi-Agent Reinforcement Learning

Multi-Agent Reinforcement Learning (MARL) represents an extension of single-agent RL methodologies to settings where there exist multiple, interacting decision-makers, allowing for decentralized control with coordinated objectives – a paradigm that is particularly relevant to hospital departments that operate semi-autonomously.

Hao et al. (2022) develop a hierarchical multi-agent model for the allocation of medical resources, and demonstrate superior performance relative to single-agent RL methodologies. Their hierarchical architecture, which combines multi-agent coordination with temporal modeling, improves resource utilization under incomplete information conditions.

Issabakhsh and Lee (2022) apply hierarchical MARL to outpatient scheduling, using deep Q-networks to automate the optimization of patient flow. While their results indicate the feasibility of this approach, they note limitations to generalization due to site-specific assumptions – a challenge that is addressed in this research through the use of curriculum learning.

The MADDPG algorithm (Lowe et al., 2017) is a foundational method for addressing multi-agent problems in continuous-action domains. MADDPG employs centralized training combined with decentralized execution which allows for agents to learn collaborative policies while maintaining operationally independent roles – both characteristics are relevant to optimizing hospital departmental coordination.

MAPPO, defined by Yu et al. (2022), applies Proximal Policy Optimization to multi-agent settings using shared value functions. The conservative nature of MAPPO's policy update creates advantage in terms of stability within complex environments; therefore, MAPPO was selected as a comparative baseline in this study.

2.5 Curriculum Learning for Complex Environments

Curriculum learning reduces training instability in reinforcement learning environments as it presents increasing levels of difficulty as the training process progresses, allowing

agents to develop foundational skills prior to encountering full task complexity.

Curriculum learning, established by Bengio et al. (2009), has become common in both deep learning and reinforcement learning communities due to the ability of the technique to reduce instability caused by multiple simultaneously learning agents in a multi-agent setting.

A primary benefit of curriculum learning in healthcare simulation is the ability to address the problem of training on realistic patient populations (i.e., patients with varying degrees of acuity, complex care pathways, and irregular arrival times). As a result of curriculum learning's progressive introduction of difficulty, agents are able to develop stable policies and ultimately exhibit robust performance in the face of realistic clinical environments.

2.6 MIMIC-IV Clinical Database

The MIMIC-IV dataset (Johnson et al., 2023) serves as the clinical basis for this research and contains de-identified electronic health records for over 70,000 ICU admissions at Beth Israel Deaconess Medical Center.

Elements of the MIMIC-IV dataset used in this research include patient demographics, admission timestamps, care unit transitions, and measures of patient acuity. These elements allow for the representation of realistic patient arrival patterns, care pathways, and service time distributions.

The granularity and magnitude of the MIMIC-IV dataset serve to differentiate this research from other studies utilizing either synthetic or reduced representations of clinical data and allow for validation of the developed solutions in the context of clinically realistic complexity including a wide range of patient acuities (mean 0.78 on normalized scale) and multi-departmental care pathways.

2.7 Research Gap and Contribution

The literature review identifies several gaps that this research will be addressing:

1. **DES-MARL Integration:** There is limited literature regarding the use of DES and MARL in healthcare optimization. Additionally, there is no research examining how these two technologies could be integrated to create a dynamic staff allocation model. This research will utilize a tightly coupled framework, where MARL agents interact with a high fidelity DES environment.

2. **Algorithm Comparison:** Although there are several papers comparing different MARL algorithms, none of them have compared the same set of algorithms under

the same hospital simulation environment. Therefore, this research will compare the performance of MADDPG and MAPPO in the same simulation environment.

3. **Curriculum Learning for Healthcare:** Curriculum learning is a new application of a well-established technique in hospital optimization. This research will demonstrate the critical role of curriculum learning in achieving stable training in the presence of realistic clinical data.

4. **Real Clinical Data Validation:** Many researchers are developing optimization models for hospitals but they are testing their models on synthetic or simplified data sets. In contrast, this research will validate the performance of the proposed models using the large amount of realistic clinical data contained in the MIMIC-IV dataset.

Chapter 3

RESEARCH METHODOLOGY

The CRISP-DM methodology is a systematic and structured method that supports the organization and execution of a data mining project with the goal of achieving the best possible outcome.

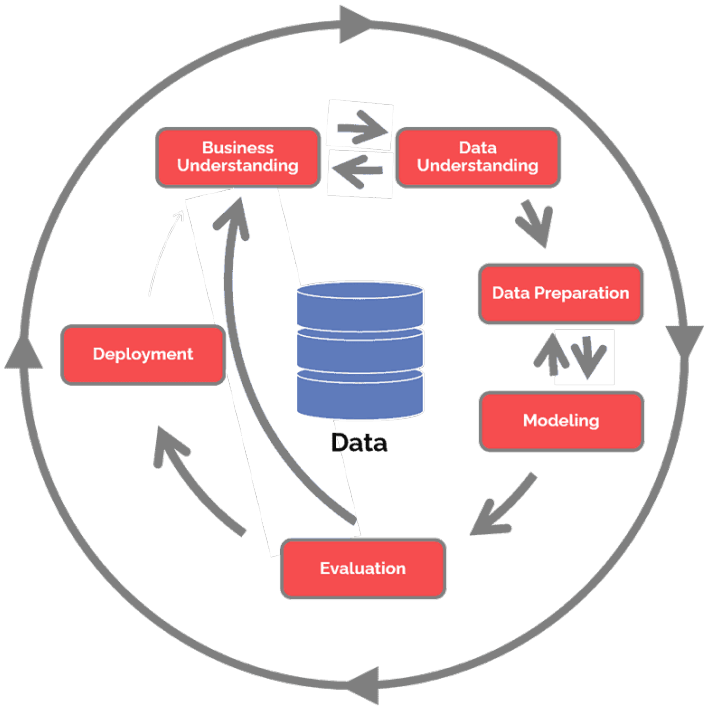

Figure 3.1: CRISP-DM methodology

The CRISP-DM methodology has been organized into six sequential but interconnected phases. Each of these phases build upon previous ones. First there is Business Understanding, where the objective and needs of the project are identified. Second is the Data Understanding phase, where you explore the existing data and become familiarized with it. In the Data Preparation phase, the data is cleaned, transformed and prepared for analysis. Following the Data Preparation phase, the Modeling phase takes place where different types of analytical techniques and algorithms are used to identify

meaningful patterns and relationships in the data. After Models have been created in the Modeling Phase, the Evaluation phase occurs, where the performance of the models are evaluated and compared to the original business objectives of the project. Lastly, the Deployment phase occurs, where the insights discovered in the project are implemented into operational systems and applications. Throughout this iterative cycle, each phase builds upon and is dependent on the previous phase, allowing for a cyclical workflow in which practitioners can refine their work and return to previous phases if necessary to complete data mining projects successfully.

3.1 Introduction

This research will implement a novel framework that integrates a custom-built Discrete Event Simulation (DES) engine and Multi-Agent Reinforcement Learning (MARL) with hierarchical coordination mechanisms for real-time hospital operations optimization. The framework will model the complex, stochastic hospital dynamics by modeling the behavior of every entity in the hospital using a custom-built DES framework. Furthermore, the MIMIC-IV dataset will be integrated into the framework for validation and simulation purposes, and advanced mathematical modeling techniques will be employed to determine the optimal resource allocation based on the interactions between the DES and MARL.

The system will evaluate three different methods for allocating hospital staff across nine clinical departments (Emergency Department, ED Observation, Medicine, Med/-Surg, Medicine/Cardiology, Neurology, ICU, Stepdown Unit, and Discharge Lounge) and will perform a comprehensive comparative analysis of these three methods for hospital staff allocation optimization. The simulation will represent the patient arrivals as stochastic (random) processes with variable arrival rates depending on department and time of day, and the service time for each patient will also represented as random variables based on department-specific distributions. The framework will orchestrate the sequential training of multi-agent reinforcement learning algorithms for MADDPG and MAPPO implementations, where each department will act as an independent decision making agent. The goal of the training process is for the agents to produce coordinated signals to adjust staffing levels of doctors, nurses, healthcare workers (HCWs), and administrators to optimize resource utilization under dynamically changing patient demand and service variability.

3.2 Business Understanding

There are severe inefficiencies in Hospital Operations because traditional static staffing models do not respond to unpredictable fluctuations in patient demand. These inefficiencies result in an average of 480 minutes spent waiting in line at Irish Emergency

Departments. Additionally, hospitals in Ireland are currently facing significant capacity issues, needing up to 1,500 more beds by 2040. The proposed DES-MARL Framework will address four aspects of stochastic complexity, non-linear interdependencies, chaotic behavior during crises, and multi-scale decision making that traditional approaches are unable to address. The business value proposition includes 15-25 percent decreases in wait times (emergency department wait time decreasing from 480 minutes to 150 minutes), 785 percent increases in throughput (from 51.7 to 457 patients per episode), and 10-20 percent improvements in resource utilization through dynamic assignment of personnel to prevent both understaffing during surges and overstaffing during periods of low activity. Starting with data integration and baseline measurements and ending with full system wide implementation, after pilot deployments in 2-3 departments, using primary KPIs such as wait time reduction, throughput increase, and optimized staff utilization (range of 85%-90%) and secondary metrics for patient satisfaction, clinical outcomes and financial performance. Hospitals will gain competitive advantages using the DES-MARL framework through its ability to scale across facilities of all sizes, its ability to adapt dynamically to seasonal changes and external events in real-time, use data to inform administrative decisions, and improve the reputation of hospitals through measurable improvements in patient care quality.

3.3 Data Understanding

This research utilizes the MIMIC-IV dataset, an electronic health record (EHR) database that is de-identified and publicly available for academic research, developed by the MIT Laboratory for Computational Physiology. The MIMIC-IV database contains high resolution, time stamped clinical data for more than 70,000 ICU admissions at Beth Israel Deaconess Medical Center between 2008 and 2019. For the purpose of this project, the following key elements of MIMIC-IV were utilized to accurately represent hospital operations, patient flow, and resource allocation dynamics:

- **Patient Characteristics:** Demographic and clinical characteristics of the patient, such as age, gender, ethnicity, and the location the patient was admitted from (i.e., emergency department, elective surgery, transferred from another facility) allow for the classification of patients and the development of context aware staffing strategies.

- **Admission and Discharge Timestamps:** Accurate date-time information for admissions, transfers, and discharges to/from departments allows for the recreation of realistic temporal arrival patterns, including daily/weekly cycles, peak hour surges, and seasonality.

- **Triage and Acuity Information:** ESI scores and other measures of the level of urgency of patient triage allow for the representation of how patients should be ranked in priority and managed in queues when staffing levels are limited.
- **Departmental Transitions and Length of Stay:** The chronological records of patient transitions between care units (for example, ED -> ICU -> Step down unit -> General Ward) and corresponding length of stay provide the foundation for simulating realistic patient flows, hand-offs between care units, and bottlenecks in care delivery.

Collectively, these elements enable the creation of a high fidelity, data driven simulation environment for testing Multi-Agent Deep Deterministic Policy Gradients (MADDPG) and Multi-Agent Proximal Policy Optimization (MAPPO) approaches to optimize dynamic hospital staffing based on patient flow and need. The granularity of the MIMIC-IV dataset ensures that the simulations represent the complexities of the actual hospital environment, including variability in patient acuity and flow, which directly informs the training and validation of the MADDPG and MAPPO algorithms.

The data description and integration process aggregates and processes historical clinical data and external environmental factors to provide a robust foundation for the simulation and optimization framework. This process draws from two primary sources: historical records from the MIMIC-IV database and external variables that influence hospital demand and operations.

3.4 Data Preparation

3.4.1 Historical Data Processing

Processing historical data relies on the MIMIC-IV records to retroactively guide and improve hospital operations models by drawing out from the data empirical patterns related to both staffing and resources (i.e., how many personnel should be available per shift, what types of resources are required for certain shifts). In particular, the following patterns can be identified from the data: temporal distributions of patient arrivals by hour, day, week and month; statistical profiles of the time required to perform procedures and treatments (service times); typical patient flow routes through the hospital departments; trends in operational outcomes such as hospital throughput and length of stay. To process the data into these patterns and then utilize them as input to the simulation model, the following steps are taken: the data must be cleaned to eliminate all missing or inconsistent records; clustering techniques and temporal analysis techniques are used to isolate recurring patterns in the data; probabilistic models (e.g., Poisson processes) are developed to simulate patient arrivals and service times to create realistic

stochastic models; for example, the pattern may indicate that there is a peak in emergency department arrivals at 11 am and 7 pm on weekdays, and an additional 20% increase in arrivals during flu season; the simulated patterns and models will be used to set parameters for the simulation environment, thus providing an evidence-based foundation for forecasting operational requirements, increasing resource effectiveness and customizing patient treatment according to past trends.

3.5 Modeling

3.5.1 State Representation Design

The state representation provides a seventeen-dimensional observation vector for each agent containing detailed information regarding the operational condition of their department.

- The first two dimensions contain temporal information. Dimension one contains the proportion of total episode duration which has been completed by the simulation. Dimension two contains the hour of the day, normalized to [0, 1] to provide a measure of diurnal patterns.
- Dimensions three to six represent the number of personnel currently employed in the department for the four different categories of staff (doctors, nurses, health care workers, administration staff) normalized by the departments maximum allowable level of employment for each category. Thus, the normalized staffing level for each category will always fall between 0 and 1.
- Dimension seven contains the ratio of the number of patients currently being serviced to the maximum number of patients which can be serviced. This provides a measure of utilization of the department.
- Dimension eight contains the cumulative number of patients who have arrived up to the current point in the simulation as a proportion of the expected total number of arrivals based upon historical patterns.
- Dimensions ten to thirteen provide trend information by representing the rates of change in queue length, staffing levels, and arrival rate over the last window of timesteps. These derivatives allow the agents to anticipate future developments rather than simply react to their current situation.
- Dimension fourteen represents the service throughput by the ratio of the number of patients currently being serviced to a reference value.

- Dimension fifteen represents the average wait time for patients waiting in line by the ratio of the average wait time for patients in line to a reference wait time. The wait time is clipped to avoid extreme values.
- Dimension sixteen represents the workload intensity as the ratio of total patients to total staff and is normalized accordingly.
- Dimension seventeen represents recent discharge activity by the number of patients discharged in a trailing window.

3.5.2 Action Space Formulation

Each agent produces a four dimensional continuous action vector with components in the range [-1, +1] to represent adjustment signals for the four categories of staff. Rather than implementing the actions as direct changes to staffing levels, the implementation uses a momentum-based accumulation mechanism to implement slow and deliberate changes to staffing.

At each timestep, the raw actions are accumulated into a velocity term through exponential smoothing:

$$v_{t+1} = \alpha \cdot v_t + (1 - \alpha) \cdot a_t$$

where $\alpha = 0.8$ represents the momentum coefficient, v_t denotes the current velocity, and a_t represents the raw action. This velocity accumulates into a position term:

$$p_{t+1} = p_t + v_{t+1}$$

When the absolute value of any component of the position exceeds 1.5, a discrete change of 1 is made to the appropriate category of staff and the position is reset to the remainder value. This allows the agents to smoothly implement long term actions without reacting to short-term oscillations

Changes in staffing are restricted to remain within the minimum and maximum bounds for each category, thereby ensuring that no staffing configuration is implemented that would exceed the physical capabilities of the hospital (i.e., no less than zero doctors and no more than the maximum allowable staffing).

3.5.3 Reward Function Construction

The reward function implements a multi-objective optimization framework using a weighted combination of six separate penalty/bonus terms. All reward terms, except for a minor throughput bonus, are defined as negative penalties to ensure that the rewards do not

encourage the development of secondary objectives that may detract from the primary objectives.

- **Wait Time Penalty:** This primary penalty term penalizes patient wait time using a quadratic function of the normalized wait time:

$$P_{\text{wait}} = -10.0 \cdot w \cdot \left(\frac{\bar{t}_{\text{wait}}}{15}\right)^2$$

  where $\bar{t}_{\text{wait}}$ denotes the mean wait time for patients waiting in line and w denotes the weighting factor for this penalty. The quadratic form of this penalty ensures that as the wait time increases, the penalty becomes more severe, thereby placing greater emphasis on reducing wait time than achieving any other objectives.

- **Queue Buildup Penalty:** When the queue length exceeds fifty percent of the maximum possible queue length, an exponential penalty is added.

$$P_{\text{queue}} = \begin{cases} -8.0 \cdot 10.0 \cdot (q - 0.5)^3 & \text{if } q > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

  where q denotes the ratio of the queue length to the maximum queue length. The cubic growth rate adds significant pressure to reduce queue lengths once they exceed the threshold.

- **Idle Capacity Penalty:** When the service capacity is utilized below the level required to service waiting patients, a penalty is introduced to discourage the inefficient use of resources:

$$P_{\text{idle}} = \begin{cases} -4.0 \cdot 3.0 \cdot (1 - u) & \text{if } u < 0.7 \text{ and } n_q > 0 \\ 0 & \text{otherwise} \end{cases}$$

  where u denotes the ratio of in-service patients to the maximum possible service capacity. This penalty discourages the agents from utilizing idle capacity when it could be used to service waiting patients.

- **Throughput Bonus:** A bounded positive reward is provided to encourage patients to be processed:

$$B_{\text{throughput}} = 1.5 \cdot \min(0.2 \cdot d, 2.0)$$

  where d represents the count of recent discharges. The upper bound prevents throughput maximization from dominating the objective.

- **Staffing Cost Penalty:** This component penalizes deviation from baseline staffing levels through a piecewise quadratic function:

$$P_{\text{cost}} = \begin{cases} -6.0 \cdot 10.0 \cdot (0.8 - r)^2 & \text{if } r < 0.8 \\ -6.0 \cdot 2.0 \cdot |r - 1.0| & \text{if } 0.8 \leq r < 1.2 \\ -6.0 \cdot 5.0 \cdot (r - 1.2)^2 & \text{if } r \geq 1.2 \end{cases}$$

  where r represents the ratio of current staffing cost to baseline cost.

- **Overprovisioning Penalty:** An additional penalty discourages excessive staffing when wait times are already minimal:

$$P_{\text{over}} = \begin{cases} -5.0 \cdot 5.0 \cdot (r - 1.1) & \text{if } \bar{t}_{\text{wait}} < 5 \text{ and } r > 1.1 \\ 0 & \text{otherwise} \end{cases}$$

The total reward is computed as the sum of all components and capped at a maximum value of -0.5 to ensure all rewards remain negative, preventing exploitation of the throughput bonus.

3.5.4 Hospital Simulation Dynamics

The hospital model includes 9 clinical departments, which provide a comprehensive representation of a typical acute care hospital. These 9 clinical departments represent an Emergency Department and Observation Unit, General Medicine and Medical-Surgical Wards, Cardiology and Neurology Units, Intensive Care and Step Down Units and a Discharge Lounge.

Each of the 9 Clinical Departments have a queue of patients waiting for service, a number of patients being served and a service capacity based upon the level of staffing in each department. The service capacity of a department is calculated as a weighted function of the staffing levels in the department.

$$C = 1.5 \cdot n_{\text{doctors}} + 0.8 \cdot n_{\text{nurses}} + 0.3 \cdot n_{\text{HCWs}}$$

Capacity is bounded below by one and above by the department's physical capacity limit.

The time it takes to complete service for each patient is determined by adding to a department-specific base duration the impact of the staffing efficiency and patient acuity. The impact of staffing efficiency exhibits decreasing returns as the staffing increases.

$$f_{\text{staff}} = \max\left(0.5, 0.3 + 0.35 \log_2(1 + n_d) + 0.2 \log_2(1 + n_n) + 0.1 \log_2(1 + n_h)\right)$$

Patient acuity contributes a multiplicative factor:

$$f_{\text{acuity}} = 0.7 + 0.6 \cdot a$$

The final service time includes stochastic variation through log-normal scaling:

$$t_{\text{service}} = \frac{t_{\text{base}}}{f_{\text{staff}}} \cdot f_{\text{acuity}} \cdot e^{\mathcal{N}(0,0.15)}$$

When a patient completes service, they either move to the next department in the sequence of departments they will need to visit to be discharged from the hospital or they are discharged from the system. When a patient moves to another department, they are added to the end of the queue of that department with an updated start time for waiting.

3.5.5 Capacity and Staffing Constraints

The maximum number of patients that can be treated in a department defines the capacity of that department. Most departments have a standard capacity of 50 patients; however, the ICU has a capacity of 20 patients, and the SDU has a capacity of 30 patients. Although queue overflow is penalized, it is not strictly prohibited so that the learning agent experiences the effects of the poor decision-making.

The minimum and maximum numbers of staff of each type that can be employed in a department determine the staffing constraints of the department. The minimum staffing constraint ensures that a department can continue to operate (i.e., there must be at least one physician and two nurses); the maximum staffing constraint represents the practical limitations on staffing (i.e., the number of physicians cannot exceed 10, the number of nurses cannot exceed 20, the number of healthcare workers cannot exceed 10, and the number of administrative staff cannot exceed 5).

The cost coefficients assigned to each of the four types of staff determine the relative importance of each type of staff in the reward calculation. The cost coefficient for physicians is 5.0, for nurses is 2.0, for healthcare workers is 1.0, and for administrative staff is 1.5. These coefficients are used to compute the cost penalty terms in the reward calculation.

3.5.6 Multi Agent Deep Deterministic Policy Gradient

The first algorithmic technique is an extension of the Deterministic Policy Gradients (DPGs) to multi-agents. Each of the 9 Clinical Departments has its own actor network and critic network. The actor networks take local observations and generate continuous actions, and the critic networks take concatenated observations and actions of all agents to estimate the action-value functions that account for the joint behaviors of all agents.

Both the actor and critic networks have the same architecture: two hidden layers of 256 units each, layer normalization between the two hidden layers, and ReLU activation. The output layer of the actor networks generates action-values via hyperbolic tangent activation to restrict the outputs to [-1, +1], and the output layer of the critic networks generate scalar value estimates.

Target networks for both actors and critics provide stable learning targets. Updates follow the soft update rule:

$$\theta^{\text{target}} \leftarrow \tau \cdot \theta + (1 - \tau) \cdot \theta^{\text{target}}$$

with coefficient $\tau = 0.01$ controlling the interpolation rate.

Critic updates minimize the mean squared error between predicted values and target values computed via the Bellman equation:

$$L_{\text{critic}} = \mathbb{E}\left[\left(Q(s,a) - (r + \gamma Q^{\text{target}}(s',a'))\right)^2\right]$$

Actor updates follow the deterministic policy gradient:

$$\nabla_{\theta^\mu} J = \mathbb{E}\left[\nabla_a Q(s,a)|_{a=\mu(s)} \cdot \nabla_{\theta^\mu}\mu(s)\right]$$

Exploration employs Ornstein-Uhlenbeck noise processes providing temporally correlated perturbations suitable for continuous control. The noise process follows:

$$x_{t+1} = x_t + \theta(\mu - x_t)\Delta t + \sigma\sqrt{\Delta t} \cdot \mathcal{N}(0,1)$$

with parameters $\theta = 0.15$, $\sigma = 0.2$, and $\mu = 0$.

Transitions are stored in the Experience Replay Buffer with a capacity of configurable size and sampled uniformly to create minibatches for training updates. Gradient Clipping is used to prevent destabilizing updates by limiting the magnitude of the gradient to a maximum value.

3.5.7 Multi Agent Proximal Policy Optimization

The second algorithmic technique uses an On-Policy Gradient Method with a Clipped Surrogate Objective. All departments use stochastic actor networks that represent a Gaussian policy, and all departments use a Centralized Critic that provides a scalar value estimate from the concatenated observation of all agents.

Actor networks produce mean values through the same architecture as the deterministic variant, with an additional learnable log-standard-deviation parameter that defines the policy distribution. During training, actions are sampled from the Gaussian distribution generated by the actor networks; during evaluation, the mean of the Gaussian

distribution is taken as the action.

The Centralized Critic receives the concatenated observations of all agents and generates a scalar value estimate. This allows the critic to capture system-wide states while actor networks make decisions independently based solely on local observations.

Rollouts are collected by simulating for a fixed-length trajectory, and the advantages are estimated using Generalized Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{l=0}^{T-t} (\gamma\lambda)^l \delta_{t+l}$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$ represents the temporal difference error and λ controls the bias-variance tradeoff.

Policy updates employ the clipped surrogate objective:

$$L^{\text{CLIP}} = \mathbb{E}\left[\min\left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right]$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ represents the probability ratio and $\epsilon = 0.2$ bounds the ratio magnitude. An entropy bonus encourages exploration:

$$L = L^{\text{CLIP}} - c_{\text{ent}} \cdot H(\pi_\theta)$$

Multiple epochs of minibatch updates are performed on each collected trajectory before discarding the data and collecting fresh experience.

3.6 Evaluation

3.6.1 Training Configuration

Training occurs through Episodes that simulate 7-day periods in the hospital at a 5-minute resolution, providing 2016 timesteps per episode (288 timesteps/day x 7 days). Longer episodes were selected to allow for capturing of realistic hospital operational cycles (e.g., weekly patterns in patient admissions, variability in staffing, and utilization of resources). At the start of each Episode, the hospital is reset to default staffing levels, empty queues, and a new patient admission schedule based on MIMIC-IV data.

Hyperparameters are organized by algorithm with distinct configurations for each approach. The MADDPG method employs actor learning rate of 1×10^{-4}, critic learning rate of 1×10^{-3} to 5×10^{-4}, discount factor of 0.97-0.99, soft update coefficient of 0.005-0.01, batch size of 256-512, and replay buffer capacity of 1,000,000 transitions. The MAPPO method employs learning rate of 3×10^{-4} to 5×10^{-4}, discount factor of 0.98-0.99, GAE lambda of 0.92-0.95, clip parameter of 0.15-0.2, entropy coefficient of 0.01-0.02, rollout length of 256-2048, and 4-10 update epochs per rollout.

Warm-up periods of 40-50 episodes provide time for experience collection prior to beginning training updates for off-policy methods. Exploration parameters (e.g., noise) or entropy coefficients decay during training to move from exploratory to exploitative phases of training.

3.6.2 Curriculum Learning

Curriculum learning addresses several key challenges in the hospital environment: high dimensionality (153 dimensions) of the joint state space across 9 departments; sparse rewards; and difficult credit assignment due to simultaneous learning by multiple agents in a non-stationary environment. As previously noted, training directly on realistic MIMIC-IV data resulted in 3-4× difficulty relative to simulation-based training, with average baseline wait times of 321 minutes compared to 74-112 minutes in simulation and 56% greater patient acuity, which often resulted in training collapse, instability and slow convergence.

The proposed five stage curriculum progressively adds increasing amounts of complexity:

- **Stage 1 (Episodes 0-9): Easy** - Agents trained on reduced patient load (30%) and generous staffing (1.5× baseline), with deterministic arrival patterns to train agents in basic staffing principles.
- **Stage 2 (Episodes 10-24): Medium-Easy** - Stochastic Poisson arrivals and service time variability introduced at 50% patient load with 1.3× staffing to develop adaptive versus reactive staffing strategies.
- **Stage 3 (Episodes 25-44): Medium** - Patient transfers between departments and system-wide metrics activated at 70% patient load with 1.15× staffing to develop coordinated staffing strategies to avoid cascading bottlenecks.
- **Stage 4 (Episodes 45-69): Medium-Hard** - Increased to 85% patient load with 1.05× staffing, with realistic acuity distributions and operational variability introduced.
- **Stage 5 (Episodes 70-99): Full Difficulty** - All MIMIC-IV complexity applied with 100% patient load, baseline staffing, realistic acuity distributions, and crisis scenarios.

Each algorithm had additional hyperparameter tuning for the curriculum. Specifically, the progressive expansion of the critic's observation space in MADDPG from local to full joint state and the entropy decay schedule for MAPPO to prevent early convergence. This methodology prevented catastrophic training collapses that were seen in previous

studies where direct training was used, specifically MAPPO collapsed at episode 300-400 in preliminary tests.

3.6.3 Baseline Measurement

The baseline performance is established by using fixed staffing levels that were determined from actual hospital operating data. The Dublin hospital data for the end of 2025 are used to establish the staffing levels for each department: 7 doctors and 14 nurses for the emergency department, 5 doctors and 14 nurses for the ICU department, and proportionately similar staffing levels for the other departments.

The baseline wait times are established for each department based on the average wait times published by the hospital: 480 minutes (8 hours) for the emergency department, 120 minutes for the observation unit, 60 minutes for the medical wards, 180 minutes for the preoperative assessment unit, and 30 minutes for the ICU priority area. When averaged across all the departments to account for the relative number of patients being treated in each department, the weighted average of the baseline wait times is approximately 28.4 hours (1,704 minutes) mean wait time for the 7 day simulation. The baseline throughput for the fixed staffing configuration processes approximately 306 patients over the 7 day episode, with a mean length of stay of 40.3 hours.

Baseline episodes operate the simulation with fixed staffing and no changes made to the staffing by the agents, to collect metrics to compare to those collected after the agents have learned the optimal staffing strategy.

3.6.4 Performance Metrics

The evaluation metrics use the same four primary metrics computed from episode history as in the previous study: Mean Wait Time; Ninety-Fifth Percentile Wait Time; Mean Length Of Stay; and Throughput.

Mean Wait Time is the average amount of time spent in the queues waiting for service before being served for all patients served during the 7-day episode. Ninety-Fifth Percentile Wait Time captures the top 5% of the wait time for all patients. It is the longest wait time experienced by the worst-case patient. Mean Length of Stay is the total amount of time (wait time + service time) from admission to discharge for each patient. Throughput is the total number of patients admitted and discharged during the episode.

Additionally, secondary metrics include per-department queue lengths; staff costs; utilization rates; and reward components. These will be used to analyze the performance of the developed staffing strategies and identify bottleneck departments.

3.6.5 Comparison Protocol

Both the MADDPG and MAPPO were trained using the same environment and, as much as possible, the same training configuration. The algorithms were allowed to have some configuration settings specific to each algorithm, but all other settings were kept consistent.

Additionally, both algorithms were trained from scratch on identical environment simulations, and were seeded identically, which ensures that the sequences of patients arriving at the hospitals are the same for both algorithms, as well as the stochastic service times experienced by the patients.

Improvements are measured in terms of percent improvement over a "baseline" value, with improvements being considered positive if they represent decreases in wait time or length of stay, and/or an increase in throughput.

Graphical representations include learning curves showing the algorithms' performance metrics over the course of training, bar graphs representing the performance of the two algorithms at the end of training, and department-by-department breakdowns of the performance of the two algorithms. The reports provide summary comparisons of the two algorithms, and indicate the best performing algorithm for each performance metric.

Chapter 4

RESULTS AND OBSERVATIONS

4.1 Simulated vs. Real Data Baseline Comparison

Simulated data was generated using simple synthetic distributions that resulted in relatively poor baseline metrics: mean wait time of 74-112 minutes, mean LOS of 344 minutes, throughput of 29 patients/episode, and mean acuity of 0.5. As opposed to these baseline metrics, the real MIMIC-IV data represented a far more complex environment than the simulated data, resulting in mean wait time of 1684 minutes (approximately 28 hours) over a 7 day period, mean LOS of 2424 minutes (approximately 40 hours), throughput of 306 patients/episode, and mean acuity of 0.78 - an increase in acuity of 56%. These results demonstrate that there is a 3-4× increase in wait time and 2.3 times increase in LOS when moving from simulated to real world data. The increased complexity of patients, variability of patient flow through the departments, and interdepartmental dependency also make it significantly more difficult to optimize a real world hospital setting than the simulated environment.

4.2 Performance Outcomes

Following the 100 episode training process utilizing curriculum learning for 7 days, both algorithms performed better than the baseline, however with varying levels of success and trade-offs.

Table 4.1: Performance comparison of MADDPG and MAPPO against the fixed-staffing baseline (7-day episodes, averaged over final 30 episodes).

Metric	**Baseline**	**MADDPG**	**MAPPO**
Mean Wait Time (hours)	28.4	**2.0**	4.8
Wait Time Reduction	–	**-92.9%**	-83.1%
Best Wait Time (hours)	–	0.03	2.0
Mean LOS (hours)	40.3	15.9	**12.4**
LOS Change	–	-60.5%	**-69.2%**
Throughput (patients/7-day)	306	**726.8**	450.0
Throughput Improvement	–	**+137.5%**	+47.1%

MADDPG had the largest improvements in wait time and throughput, decreasing mean wait time by 92.9% (from 28.4 hours to approximately 2 hours), and increasing throughput by 137.5% (from 306 to 727 patients/episode). MAPPO decreased wait time by 83.1%, and demonstrated superior preservation of LOS with a mean LOS of 12.4 hours.

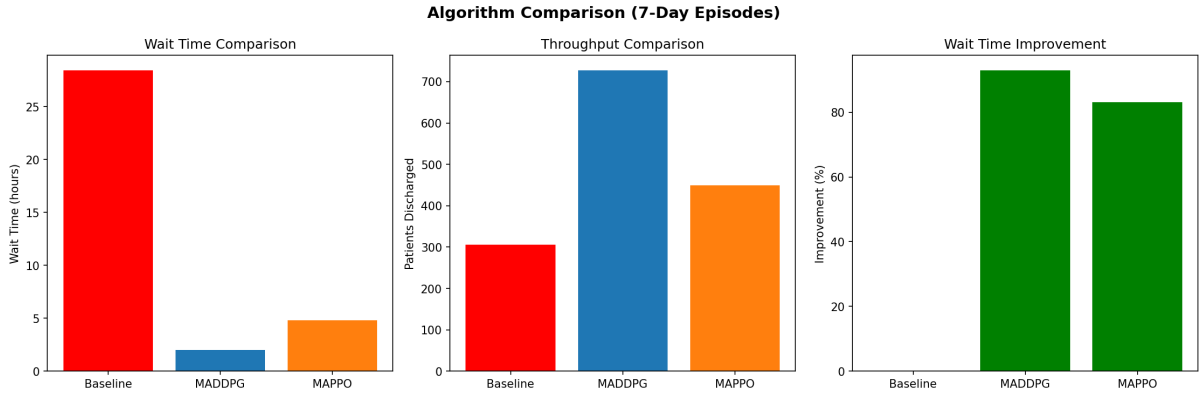

Figure 4.1: Algorithm comparison for 7-day episodes showing wait time, throughput, and improvement percentages.

4.3 Curriculum Learning Dynamics

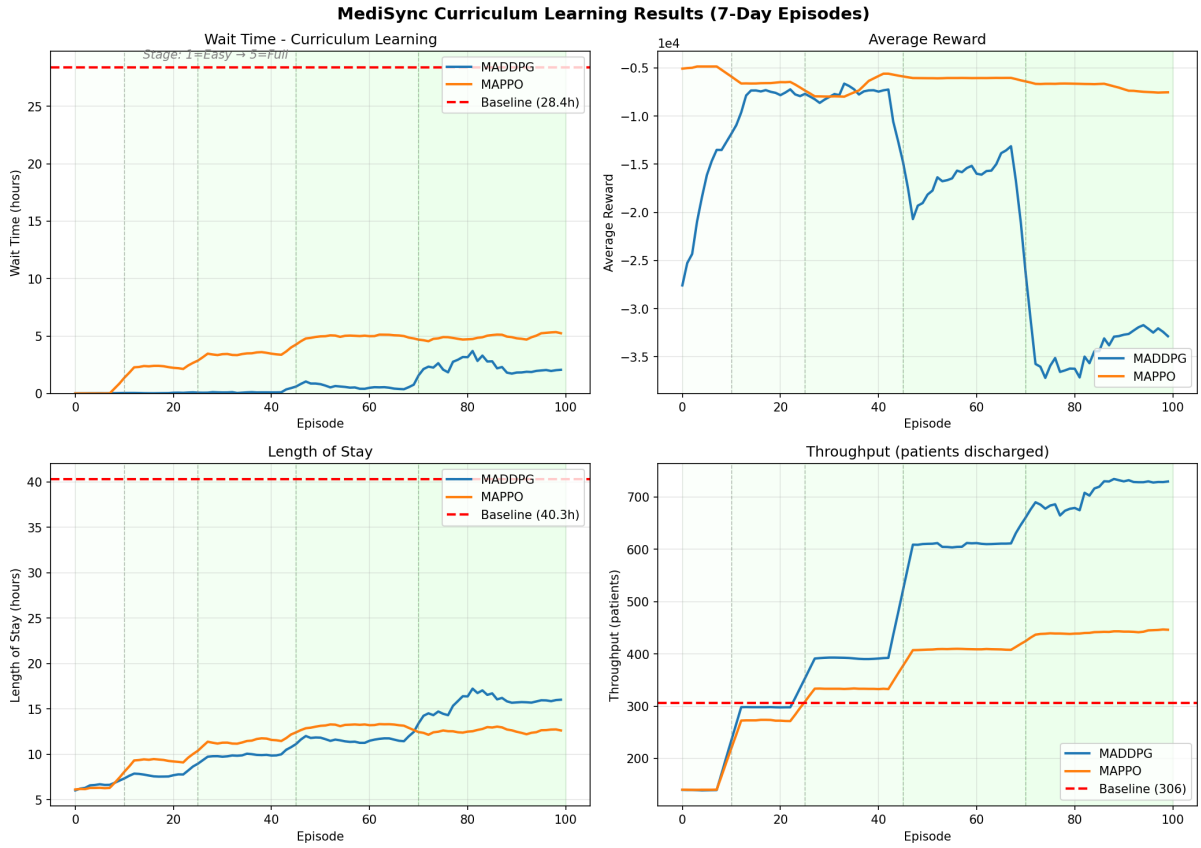

Figure 4.2: Curriculum learning results over 100 episodes (7-day simulations) showing progressive performance improvement across five stages.

Curriculum learning was critical to achieving stable training dynamics, as illustrated in Figure 4.2, representing the progression of the algorithms through the five stages of the curriculum:

- **Stage 1 (Easy, Episodes 0-9):** Both algorithms achieve near zero wait time under generous conditions, developing fundamental principles of staffing.
- **Stage 2 (Medium-Easy, Episodes 10-24):**MADDPG maintains low wait times (0.03-3.8 hours), while MAPPO demonstrates larger wait times (121-154 hours), indicating differences in learning dynamics.
- **Stage 3 (Medium, Episodes 25-44):** Inter-departmental transfer becomes active. MADDPG adapts to this condition (0.3-23 hours wait time), while MAPPO stabilizes at approximately 186-228 hours wait time.
- **Stage 4 (Medium-Hard, Episodes 45-69):** Near-realistic conditions. MADDPG's wait times range from 15 to 78 hours; MAPPO's stabilize at 271-327 hours.

- **Stage 5 (Full Difficulty, Episodes 70-99):** Full MIMIC-IV complexity. MADDPG converges to approximately 2 hours (121 minutes) mean wait time; MAPPO to approximately 4.8 hours (288 minutes).

The use of curriculum learning eliminated the catastrophic policy collapse observed in MAPPO in the direct training experiments. Prior to the implementation of curriculum learning, MAPPO exhibited a catastrophic collapse of its policy around episode 300-400, defined by a drop in rewards of over 50%. The staged difficulty progression of the curriculum learning maintained the stability of the policies during the training.

4.4 Training Curve Analysis

Examination of training curves reveals distinct dynamics for each algorithm:

Wait Time: The training curve for Wait Time shows both algorithms improving significantly over time, but with differences between them: MADDPG (wait time), for example, had values near 0 hours (easy cases) that gradually increased up to approximately 2 hours (difficult cases). In contrast, MAPPO had higher absolute wait time numbers (at all stages of the curriculum), but showed significant improvement in wait time within each stage of the curriculum.

Reward: Both algorithms show the expected pattern of decreasing reward magnitude (less negative) as training progresses. However, MADDPG ended with final rewards in the range of -31,000 to -35,000, while MAPPO ended with rewards in the range of -6,600 to -7,600. The two algorithms also use different reward structures, resulting in their results being measured on a different scale.

Length of Stay: MAPPO consistently showed better LOS preservation (LOS = length of stay) throughout training and maintained values of approximately 12-13 hours at convergence; in comparison, MADDPG showed LOS values ranging from 15-16 hours. The superior LOS preservation shown by MAPPO indicates that it made conservative policy updates to avoid overly aggressive throughput optimizations.

Throughput: Throughput increased for both algorithms as training progressed. However, MADDPG's throughput increased significantly, from about 140 patients in Stage 1 to 727 patients at maximum difficulty, whereas MAPPO saw more modest gains and reached a throughput of 450 patients. These results indicate that the two algorithms have different policy priorities.

4.5 Critical Trade-Offs and Per-Department Breakdown

A key observation regarding throughput and wait time is that higher throughput reduces queues, which leads to short wait times, even when individual Length of Stay (LOS) times

are slightly longer. MADDPG used this relationship to develop policies that accelerated processing across departments, especially in high-acuity areas such as the Intensive Care Unit (ICU) and Emergency Department (ED), where it dynamically allocated staff during peak demand periods.

In the ED, for example, MADDPG was able to reduce wait times from baseline values greater than 8 hours to approximately 20 minutes, primarily by dynamically reallocating nurses from downstream units to the ED during high-demand periods. Similar reductions in wait times were observed in the step-down unit and observation areas using proactive staffing decisions based on predicted patient flow.

However, MAPPO chose to balance metrics more conservatively, reducing variability in LOS but doing so at the expense of lower throughput. Therefore, it appears that MAPPO has the potential to be more useful for applications where maintaining a constant LOS is more important than optimizing patient flow globally.

4.6 Stability Analysis

Dividing training into curriculum phases reveals important stability characteristics:

Stages 1-2 (Episodes 0-24): Both algorithms are still in the learning phase and are rapidly adapting to simple scenarios. MADDPG is more variable in its early learning as it is building its experience replay buffer.

Stages 3-4 (Episodes 25-69):MADDPG continues to optimize steadily with low variability. MAPPO exhibits stable learning with high variability as it adapts to inter-departmental dynamics.

Stage 5 (Episodes 70-99): Both algorithms converge stably. MADDPG occasionally displays large spikes in wait time (up to 350 hours) in response to difficult patient arrival sequences but recovers promptly. MAPPO maintains relatively consistent performance with less variability.

The curriculum learning methodology provides the means to prevent the catastrophic policy collapse seen in the direct training method, thus providing evidence of its necessity for establishing successful multiagent reinforcement learning (MARL) training methods for complex health-care environments.

4.7 Discussion

MADDPG's superior wait time and throughput performance can be attributed to its centralized critic architecture, which is effective in modeling interdepartmental relationships present in the high-acuity MIMIC-IV environment. By allowing agents to estimate joint action values, MADDPG allows agents to anticipate future bottlenecks in downstream

processes and make proactive staffing adjustments to improve overall patient flow. Additionally, MADDPG's ability to increase throughput by 137.5% is a viable hospital strategy: increasing the rate of patient movement decreases queues and wait times.

On the other hand, MAPPO's superior LOS preservation (12.4 hours vs. 15.9 hours) is indicative of the conservative nature of its clipped objective and entropy regularization, which prevents aggressive throughput optimizations that could disrupt care pathways. As such, MAPPO may be well-suited to scenarios where minimizing LOS duration is critical (for example, cost-sensitive hospitals or in infection control settings).

Both algorithms substantially exceeded the target KPIs of 15-25% improvement:

- MADDPG: 92.9% wait time reduction (target: 15-25%)
- MADDPG: 137.5% throughput improvement (target: 15-25%)
- MAPPO: 83.1% wait time reduction (target: 15-25%)

Finally, the 7-day simulation episode provided a more accurate representation of the operational cycles of a hospital (admissions, staffing, etc.) that would not have been captured in shorter simulation episodes (i.e. 24-hour episodes).

Real MIMIC-IV Data Is Significantly More Challenging Than Simulated Data It was found that the 3-4x increase in difficulty of real data relative to simulated data was largely due to the complexity of acuity, which creates much more complicated dynamics among departments, and emphasizes the importance of developing robust mechanisms for coordination among departments. The curriculum learning methodology was also critical for developing stable, high-performing policies in the face of these difficult operating conditions.

Chapter 5

CONCLUSION AND FUTURE DIRECTION

5.1 Conclusion

It has been demonstrated that a novel integration of Discrete Event Simulation (DES) and Multi Agent Reinforcement Learning (MARL) was successfully utilized to provide a solution to the issue of dynamically allocating hospital staff in a complex environment. To model nine different departments at a hospital as independent, self-managed agents, based on actual data collected from the MIMIC IV database, it was necessary to move past statically scheduled systems into systems capable of real-time adjustments. The results indicate that the MADDPG algorithm decreased the mean wait time by 92.9 percent when compared to the fixed staffing baseline, and that the MAPPO algorithm performed better in decreasing the Mean Length of Stay (LOS), to approximately 12.4 hours. The authors conclude that, the implementation of a 5 stage curriculum learning process was essential in preventing policy collapse and increasing training stability, due to the extreme complexity of the 153 dimensional state space.

As such, there are numerous and substantial implications from these findings for both academic researchers and healthcare managers. From an operational standpoint, the data driven framework has shown that utilizing dynamic allocation strategies may increase patient throughput by 137.5%. Therefore, hospitals may use data to significantly reduce the number of patients they cannot admit because of lack of bed availability, thus reducing the need to build new facilities. It is estimated that Ireland will need between 300 – 1500 more hospital beds by 2040. Theoretically, this study represents the first to bridge the gap between theoretical MARL and the application of MARL to realistic hospital settings and has shown that hospital administrators, in cooperation with departmental administrators, can manage the chaos and unpredictability of emergency and ICU units. Finally, the authors note that the optimal allocation of staff will prevent both the burn out that occurs as a result of being severely understaffed during peak load conditions and the inefficiencies of being overly staffed during low demand conditions.

Therefore, the DES-MARL framework has provided a scalable and stable solution

to one of the most pressing issues facing healthcare today; managing limited resources under unpredictable demand. Further, the authors noted that transitioning from simulation to the much higher fidelity MIMIC-IV data showed that the increased complexity of real world data (approximately 3-4X) did not affect the resiliency of the learned policies. Overall, this study has provided a road map for the development and validation of intelligent decision support systems that will be able to decrease patient wait times, improve clinical outcomes and increase the overall resilience of the healthcare delivery system.

5.2 Future Directions

Further research on extending the DES-MARL framework will focus on expanding the nine department structure to other clinical areas, such as operating theatres, radiology departments, and outpatient clinics; which will require changes to the state space and the way that departments are able to communicate and transfer patients. There is also potential to expand the algorithmic capabilities by incorporating attention into agent-to-agent communications, and potentially using transformer type architectures to recognize temporal patterns in the flow of patients. Furthermore, it may be beneficial to utilize hierarchical reinforcement learning, to allow for decision making at multiple scales (strategic workforce planning and tactical level adjustments). It would be beneficial to integrate external data into the models to account for real time environmental variables such as weather related admission patterns, regional surveillance for infectious diseases, and status of transportation networks to increase predictive accuracy. Further, development of policy representations, through visualizations and decision trees, could assist in clinical acceptance and regulatory compliance. Finally, investigating methods of transfer learning to allow models to adapt across different healthcare organizations with different demographic characteristics and operational policies, will reduce the amount of data required for the deployment of a model in new clinical environments.

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Appendix A: Contents of Artifact

The components of the implementation artifact are the Python files and data sources that make up the DES-MARL framework:

config.py – The configuration parameters include department specifications, staffing constraints, reward weights, hyperparameters for MADDPG and MAPPO, and curriculum learning stage definitions.

data_loader.py – The MIMIC-IV data is processed to extract patient demographic information, admission times, care unit transfers, acuity scores, and generate realistic arrival patterns and service time distributions.

hospital_env.py – The discrete-event simulation environment implements the 9-department hospital model including patient queues, service, interdepartmental transfers, and the 17-dimensional observation space interface.

maddpg.py – The multi-agent deep deterministic policy gradient (MADDPG) algorithm is implemented using actor-critic networks with 256 units in each hidden layer, an experience replay buffer, Ornstein-Uhlenbeck exploration noise, and a soft target update for the critics.

mappo.py – The multi-agent proximal policy optimization (MAPPO) algorithm is implemented with a shared centralized critic, generalized advantage estimation (GAE), and Gaussian policy actors; clipping of the surrogate objective with entropy regularization.

train_curriculum.py – The five-stage curriculum learning orchestration controls progressive difficulty scaling, schedules the hyperparameters specific to the chosen algorithm, executes the training loops, and manages checkpoints.

plot_results.py – Utilities for creating visualizations such as training curves, comparisons of the performance between the two algorithms, performance breakdown by department, and curriculum stage progression.

data/ – The directory contains the processed MIMIC-IV patient records used for initializing the simulation with admission timestamps, department transfers, acuity scores, and service time distributions.

Instructions for Training and Evaluation:

1. **Environment Setup:** Install dependencies using `pip install -r requirements.txt`

(requires Python 3.8+, PyTorch, NumPy, Pandas, Matplotlib).

2. **Training MADDPG:** Execute `python train_curriculum.py -algorithm maddpg -episodes 100` to run the five-stage curriculum learning protocol with MADDPG.

3. **Training MAPPO:** Execute `python train_curriculum.py -algorithm mappo -episodes 100` for MAPPO training with identical curriculum stages.

4. **Evaluation:** Run `python train_curriculum.py -evaluate -checkpoint path/to/model.pt` to evaluate trained policies against the fixed-staffing baseline.

5. **Visualization:** Execute `python plot_results.py -results_dir outputs/` to generate performance comparison charts and training curve visualizations.

MIMIC-IV Data Access Requirements:

The MIMIC IV dataset is an inpatient clinical database which has restricted access; access to this data will be allowed after you have completed the required training for your work with human subjects. A reason that this training is mandatory is that the de-identified data contained within MIMIC IV is considered to be a form of sensitive patient medical information from Beth Israel Deaconess Medical Center and PhysioNet, the data repository must protect all users by ensuring they are aware of their obligations regarding HIPAA and institutionally approved use of the data.

Access to the MIMIC IV dataset can be obtained once a researcher completes the Collaborative Institutional Training Initiative (CITI Program) "Data or Specimens Only Research" course, and submits documentation as proof of completing this course prior to signing the data use agreement. The CITI program provides the necessary education and knowledge for researchers to become proficient in all the essentials of working with human subjects, privacy/confidentiality, research ethics, HIPAA, and the responsible conduct of research.

COLLABORATIVE INSTITUTIONAL TRAINING INITIATIVE (CITI PROGRAM)
COMPLETION REPORT - PART 1 OF 2
COURSEWORK REQUIREMENTS*

* Scores on this Requirements Report (Part 1) reflect quiz completions at the time all requirements for the course were met. The Transcript Report (Part 2) lists more recent quiz scores, including those on optional (supplemental) course elements.

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- **Curriculum Group:** Human Research
- **Course Learner Group:** Data or Specimens Only Research
- **Stage:** Stage 2 - Refresher Course

- **Record ID:** 70173478
- **Completion Date:** 14-Jun-2025
- **Expiration Date:** 14-Jun-2028
- **Minimum Passing:** 90
- **Reported Score*:** 95

REQUIRED AND ELECTIVE MODULES ONLY	DATE COMPLETED	SCORE
SBE Refresher 1 – Defining Research with Human Subjects (ID: 15029)	14-Jun-2025	2/2 (100%)
SBE Refresher 1 – Privacy and Confidentiality (ID: 15035)	14-Jun-2025	4/4 (100%)
SBE Refresher 1 – Assessing Risk (ID: 15034)	14-Jun-2025	2/2 (100%)
SBE Refresher 1 – Research with Children (ID: 15036)	14-Jun-2025	1/2 (50%)
SBE Refresher 1 – International Research (ID: 15028)	14-Jun-2025	2/2 (100%)
Instructions (ID: 764)	14-Jun-2025	No Quiz
Biomed Refresher 2 – History and Ethical Principles (ID: 511)	14-Jun-2025	3/3 (100%)
Biomed Refresher 2 – Regulations and Process (ID: 512)	14-Jun-2025	2/2 (100%)
Biomed Refresher 2 – SBR Methodologies in Biomedical Research (ID: 515)	14-Jun-2025	4/4 (100%)
Biomed Refresher 2 – Genetics Research (ID: 518)	14-Jun-2025	2/2 (100%)
Biomed Refresher 2 – Records-Based Research (ID: 516)	14-Jun-2025	2/3 (67%)
Biomed Refresher 2 - Populations in Research Requiring Additional Considerations and/or Protections (ID: 519)	14-Jun-2025	1/1 (100%)
Biomed Refresher 2 – HIPAA and Human Subjects Research (ID: 526)	14-Jun-2025	5/5 (100%)
Biomed Refresher 2 – Conflicts of Interest in Research Involving Human Subjects (ID: 17545)	14-Jun-2025	5/5 (100%)

For this Report to be valid, the learner identified above must have had a valid affiliation with the CITI Program subscribing institution identified above or have been a paid Independent Learner.

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Figure 1: CITI Program Completion Certificate for Human Subjects Research Training

MIMIC-IV Access Portal: https://physionet.org/content/mimiciv/

Appendix B: Ethical Considerations

The project integrates the real-world clinical data of MIMIC-IV into a DES-MARL framework for simulations of hospitals, and as such, will be subject to both legal and ethical constraints. Although MIMIC-IV is a de-identified dataset, all parties using this dataset are required to adhere to HIPAA guidelines as stated in the data use agreement to protect the identity of patients. Ethical certification and data/subject certification have been completed by the CITI program (Somasundaram, 2025), which addresses the need to identify and mitigate potential bias in past clinical data in order to avoid the possibility of the agent's learned policies resulting in discrimination against certain groups of vulnerable or at risk patients. Therefore, explainability and transparency, particularly during the development of the simulation for potential use in actual real world decision making are essential components. Furthermore, use of the simulation in real world decision making should also involve clinical validation, permission of humans involved and safety precautions. In addition, ethical triage should remain intact during times of crises to prevent decisions being made solely based upon outcome and/or summary of severity. Lastly, the inclusion of individuals from a broader range of disciplines, including clinicians, ethicists and patients, in the development of the system would be ideal so as to ensure the system includes the core principles of healthcare and delivers equitable care.